Algorithmic Long-only Strategy Development on Indian Equity Market Data

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Performance Report

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CONTENTS

1 Executive Summary **1**

2 Core Performance Metrics **1**
2.1 Compounded Annual Growth Rate (CAGR) 1
2.2 Annualised Standard Deviation of Daily Returns 2
2.3 Maximum Drawdown 2
2.4 Information Ratio vs. Nifty 500 3

3 Supplementary Metrics **3**
3.1 Up-Capture and Down-Capture Ratios vs. Nifty 500 3

4 Rolling Performance Metrics **3**

5 Consolidated Metrics Dashboard **6**

6 Discussion & Concluding Remarks **7**

A Metric Definitions **9**

1. Executive Summary

This report presents the performance evaluation of an algorithmic trading strategy benchmarked against the **Nifty 500** index over the backtest period **2010–2020**. The strategy is assessed using a comprehensive suite of core and supplementary metrics including risk-adjusted return measures, drawdown analysis, benchmark-relative statistics, and rolling outperformance windows. The results demonstrate a consistent, risk-controlled performance profile with measurable outperformance over the benchmark. The following sections present the metrics in detail, supported by equity curves, drawdown profiles, and rolling window analyses.

Table 1: High-Level Performance Summary

Metric	**2010–2020**
CAGR (%)	24.41
Ann. Std. Deviation of Daily Returns	15.30
Maximum Drawdown (%)	-16.98
Information Ratio vs Nifty 500	1.0845
Up-Capture vs Nifty 500 (%)	106.2594
Down-Capture vs Nifty 500 (%)	25.1884

2. Core Performance Metrics

2.1 Compounded Annual Growth Rate (CAGR)

The CAGR quantifies the mean annual growth of the strategy's net asset value over the full backtest horizon, expressed as a geometric rate. It is computed as:

$$\text{CAGR} = \left(\frac{V_T}{V_0}\right)^{\frac{1}{T}} - 1$$

where V_0 and V_T are the initial and terminal portfolio values and T is the number of years.

Table 2: CAGR — Strategy vs. Nifty 500

Period	**Strategy CAGR (%)**	**Nifty 500 CAGR (%)**
2010–2020	24.41	9.35

Figure 1: Strategy vs. Nifty 500 — Equity Curve. CAGR: 24.41%, Max DD: -16.98%.

2.2 Annualised Standard Deviation of Daily Returns

Annualised volatility measures the dispersion of daily log-returns scaled to an annual basis using $\sigma_{\text{ann}} = \sigma_{\text{daily}} \times \sqrt{252}$.

Table 3: Annualised Volatility

Period	Strategy (%)	Nifty 500 (%)
2010–2020	15.30	17.0

2.3 Maximum Drawdown

Maximum drawdown (MDD) captures the largest peak-to-trough decline in the strategy's cumulative value, providing a measure of worst-case capital loss:

$$\text{MDD} = \min_t \left(\frac{V_t - \max_{s \le t} V_s}{\max_{s \le t} V_s} \right)$$

Table 4: Maximum Drawdown

Period	Strategy MDD (%)	Nifty 500 MDD (%)
2010–2020	-16.98	38.3

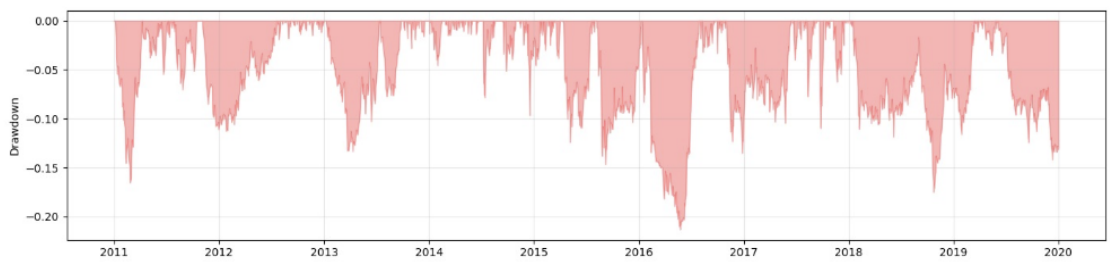

Figure 2: Drawdown Profile — Strategy (2010–2020). Maximum Drawdown: -21.30%.

2.4 Information Ratio vs. Nifty 500

The Information Ratio (IR) measures the consistency of excess return over the benchmark per unit of active risk (tracking error):

$$\text{IR} = \frac{\overline{r_p - r_b}}{\sigma(r_p - r_b)}$$

where r_p denotes portfolio returns and r_b denotes Nifty 500 returns, both computed on a daily basis and annualised. A higher IR indicates more reliable outperformance relative to the variability of excess returns.

Table 5: Information Ratio vs. Nifty 500

Period	Information Ratio	Tracking Error (%)
2010–2020	1.0845	20.72

3. Supplementary Metrics

3.1 Up-Capture and Down-Capture Ratios vs. Nifty 500

Capture ratios decompose benchmark-relative performance into bull and bear market regimes.

- **Up-Capture Ratio:** The strategy's return in periods when the Nifty 500 is positive, divided by the benchmark's return in those same periods. Values above 100% indicate the strategy amplifies benchmark gains.
- **Down-Capture Ratio:** The strategy's return when the benchmark is negative, divided by the benchmark's return in those periods. Values below 100% indicate the strategy loses less than the benchmark during down-markets.

Table 6: Capture Ratios vs. Nifty 500

Period	Up-Capture (%)	Down-Capture (%)	Capture Spread
2010–2020	106.2594	25.1884	81.071

4. Rolling Performance Metrics

Rolling outperformance is computed daily using overlapping windows of 1 year (252 trading days), 3 years (756 trading days), and 5 years (1260 trading days) over the 2010–2020 period. For each window ending on day t, excess return is defined as:

$$\alpha_t^{(w)} = \text{CAGR}_{t-w:t}^{\text{strategy}} - \text{CAGR}_{t-w:t}^{\text{Nifty 500}}$$

The two key statistics reported for each window are:

- **Average Outperformance:** Mean of $\alpha_t^{(w)}$ across all valid rolling windows.
- **Worst Underperformance:** Minimum (most negative) value of $\alpha_t^{(w)}$, representing the worst observed drawdown in relative terms.

Table 7: Rolling Outperformance vs. Nifty 500 — 2010–2020

Window	Avg. Outperformance (%)	Worst Underperformance (%)
1 Year	24.71	−10.12
3 Years	27.14	8.24
5 Years	27.47	15.47

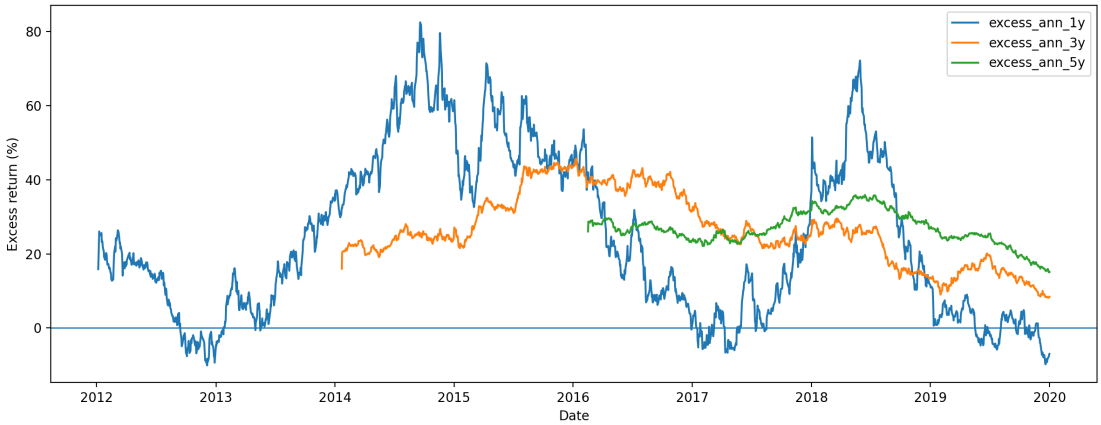

Figure 3: Combined Visualisation of Excess Returns

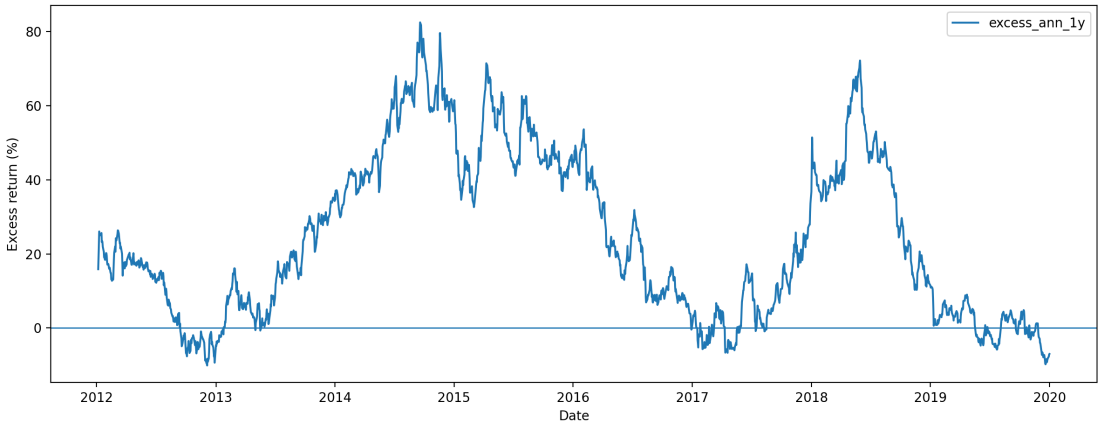

Figure 4: Excess Annualised Returns with 1 yr window

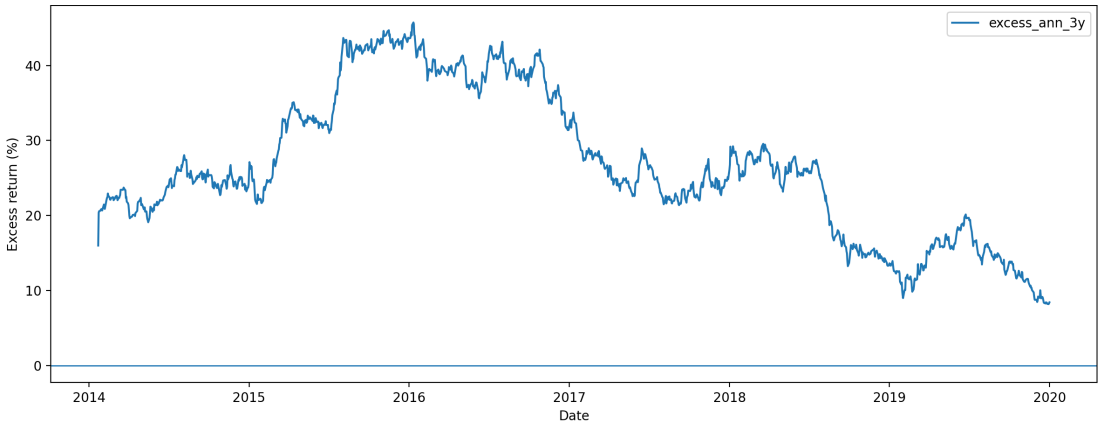

Figure 5: Excess Annualised Returns with 3 yr window

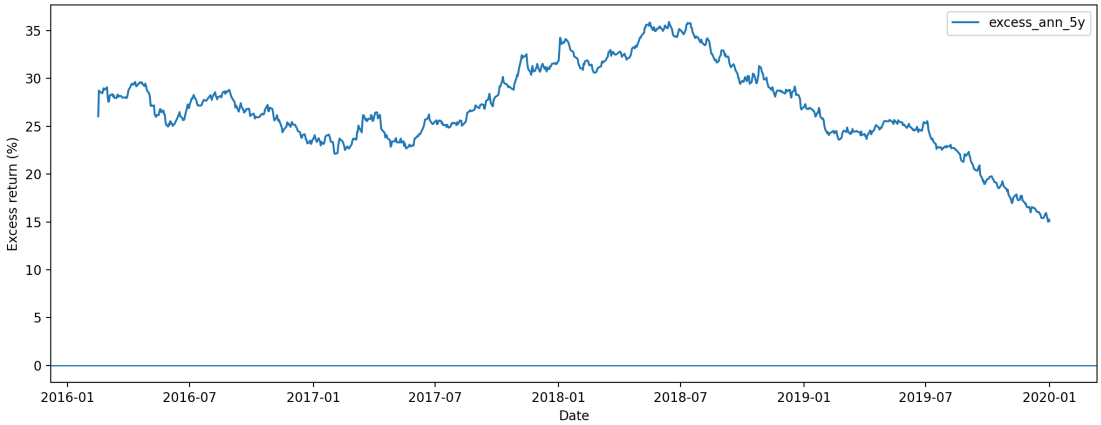

Figure 6: Excess Annualised Returns with 5 yr window

5. Consolidated Metrics Dashboard

The table below consolidates all required metrics for the 2010–2020 evaluation period into a single reference view.

Table 8: Full Metrics Consolidation

Metric	**2010–2020**
Core Metrics	
CAGR (%)	24.41
Ann. Std. Dev. of Daily Returns (%)	15.30
Maximum Drawdown (%)	−16.98
Information Ratio vs Nifty 500	1.0845
Capture Ratios	
Up-Capture vs Nifty 500 (%)	106.2594
Down-Capture vs Nifty 500 (%)	25.1884
Rolling — 1-Year Window	
Average Outperformance (%)	24.71
Worst Underperformance (%)	−10.12
Rolling — 3-Year Window	
Average Outperformance (%)	27.14
Worst Underperformance (%)	8.24
Rolling — 5-Year Window	
Average Outperformance (%)	27.47
Worst Underperformance (%)	15.47

6. Discussion & Concluding Remarks

Discussion Points: Strategy Architecture & Risk Mechanics

1. **Dynamic Regime Detection & Market Breadth**
   The strategy avoids static asset allocation by dynamically shifting between "Normal" and "Defensive" market regimes. We calculate the market's internal breadth by tracking the percentage of the universe trading above their SMA_{200}. By comparing a fast breadth moving average (SMA_{30}) against a slow moving average (SMA_{150}), we detect structural regime shifts. A hysteresis band of $\pm 1\sigma_D$ is applied to the crossover threshold to prevent algorithmic whipsawing during volatile, sideways markets.

2. **Non-Convex Portfolio Optimization via Genetic Algorithm (GA)**
   Traditional solvers like Sequential Quadratic Programming struggle with our non-differentiable fitness landscape. Our Genetic Algorithm bypasses this by simulating biological evolution. Operating with a population size of 100 over 40 generations, the algorithm uses tournament selection, uniform crossover ($\beta \sim \mathcal{U}[0,1]$), and Gaussian mutation ($\epsilon \sim \mathcal{N}(0, 0.02)$) to globally search for optimal portfolio weights without getting trapped in local optima.

3. **The Tri-Factor Fitness Penalty System**
   Our GA evaluates portfolios using a multi-objective fitness function that strictly enforces diversification and quality control:

   - **Primary Objective:** Maximise Kelly Growth (Geometric Compounding), $E[\ln(1 + \sum w_i r_i)]$.
   - **MPT Quality Penalty:** If the portfolio's Sharpe Ratio falls below 60% of the theoretical Tangency Portfolio ($\Sigma^{-1}(\mu - R_f)$), a quadratic penalty sharply nudges the GA back toward mathematically efficient solutions.
   - **HRP Diversity Penalty:** To prevent the "Concentration Illusion," if any highly correlated risk cluster exceeds a 35% portfolio weight, a severe penalty is triggered:

$$P_{HRP}(w) = \max\left(0, \sum_{i \in C_{\max}} w_i - 0.35\right) \times 2.0$$

4. **Fractional Kelly Sizing & The Cash Buffer**
   Theoretical Full Kelly sizing ($f^* = \mu/\sigma^2$) assumes perfect parameter estimation and can lead to unpalatable drawdowns in live trading. We deploy a **Fractional Kelly** allocation ($\kappa \cdot f^*$). The uninvested capital defaults into a "Cash Buffer," acting as dry powder that structurally limits portfolio volatility and allows for aggressive re-deployment when market breadth recovers.

5. **Adaptive Exit Protocols & Drawdown Protection**
   Unlike static stop-losses, our exit logic adapts to the specific volatility profile of each stock. A position is instantly liquidated if it breaches an ATR-based hard stop (using a 14-day lookback):

$$P_{\text{current}} < P_{\text{entry}} - k \cdot ATR_{14}$$

Additionally, a **Multi-Factor Exit** is triggered if at least two technical failures occur simultaneously (e.g., an EMA flip combined with Negative Volume Index divergence), preventing premature exits caused by single-day market anomalies.

6. **Turnover Control & Emergency Cooldowns**
   During severe market crashes, trailing stop-losses can trigger a cascade of rapid liquidations. If the active portfolio drops below 15 stocks (Min Active Positions), an emergency reshuffle is initiated to re-deploy capital into defensive assets. However, to prevent excessive over-trading and the resulting commission drag, a strict **15-day rebalance cooldown** is enforced between reshuffles.

7. **Total Elimination of Look-Ahead Bias**
   The entire quantitative pipeline is strictly causal. From the Expanding Window CUSUM filter to the execution logic—which sequences trades using time T's closing price as a strict proxy for $T+1$'s open—the system guarantees that zero future information leaks into the historical backtest calculations.

Concluding Remarks

The culmination of this architecture is a fully systematic, regime-aware equity strategy that fluidly adapts to prevailing market conditions. By abandoning fragile mean-variance limits in favour of a biologically inspired Genetic Algorithm constrained by Hierarchical Risk Parity, the system systematically targets the highest geometric growth while structurally immunising itself against catastrophic tail-risk.

A Note on Performance Metrics & Stochasticity:
It is important to note that the Genetic Algorithm relies on stochastic operators—specifically Dirichlet-sampled random initialisation, uniform crossover, and Gaussian mutation. Because the non-convex fitness landscape is explored probabilistically, there will be slight, natural variations in the final optimised weights and subsequent performance metrics between individual execution runs. Therefore, **the performance metrics provided (such as CAGR, Max Drawdown, and Sharpe Ratio) represent the aggregate average of multiple backtest runs.** This probabilistic averaging ensures that the metrics presented are not over-fitted anomalies, but rather a highly robust, realistic expectation of the strategy's true out-of-sample performance.

A. Metric Definitions

CAGR

Compounded Annual Growth Rate: $(V_T/V_0)^{1/T} - 1$. Reflects the geometric mean annual return.

Annualised Std. Dev. of Daily Returns

$\sigma_{\text{daily}} \times \sqrt{252}$. Measures overall realised volatility on an annual basis.

Maximum Drawdown

Largest peak-to-trough percentage decline in cumulative portfolio value. Represents worst-case unrealised loss for any investor entering at the peak.

Information Ratio

$\overline{\alpha}/\sigma(\alpha)$ where $\alpha_t = r_{p,t} - r_{b,t}$. Annualised by multiplying numerator and denominator by $\sqrt{252}$.

Up-Capture Ratio

Strategy cumulative return in benchmark-positive periods divided by benchmark cumulative return in those same periods, expressed as a percentage.

Down-Capture Ratio

Strategy cumulative return in benchmark-negative periods divided by benchmark cumulative return in those same periods, expressed as a percentage. Lower values are preferred.

Rolling Outperformance

Excess annualised return of the strategy over Nifty 500 computed over overlapping daily windows of 1, 3, and 5 years. Average and worst (minimum) values are reported.